INCH-POUND

MIL-PRF-27/44C
w/AMENDMENT 2
28 September 2020
SUPERSEDING
MIL-PRF-27/44C
w/AMENDMENT 1
26 November 2015

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 16 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.

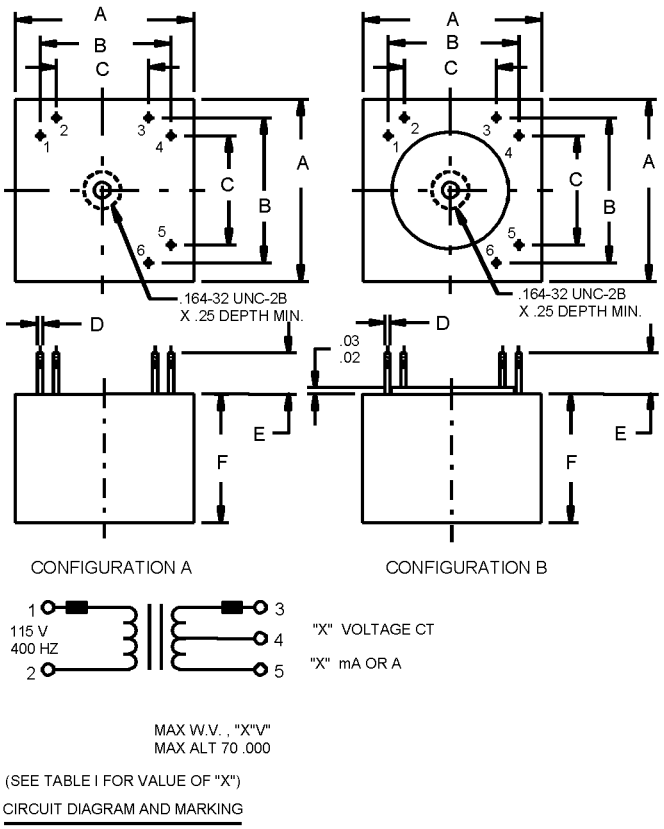


FIGURE 1. Dimensions and configurations

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LTR	Inches	mm
A	1.40 Max	35.56 Max
B	1.100	27.94
C	.900	22.86
D	.032 DIA ±.002 TYP	0.8128 DIA ±.0508
E	.13 to .25	3.302 to 6.35
F	0.97 Max	24.64 Max

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations – continued.

REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 3.4 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic or epoxy fiberglass encapsulated.

Material: Plastic or epoxy fiberglass.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C.

Terminals: Printed circuit pins – tinned brass.

Terminal strength: MIL-STD-202-211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

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No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 16 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 87 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 8 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 32°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 98˚C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

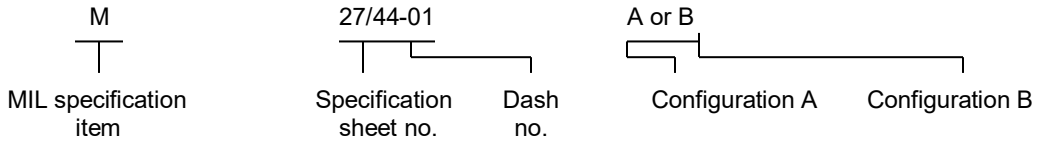
VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/44-44 shall be sufficient to grant qualification approval to M27/44-01 through M27/44-44.

Part or Identifying Number (PIN), example:



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TABLE I. Electrical ratings.

Dash No.	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	6	2.667	325
02 A or B	6.3	2.540	325
03 A or B	7	2.286	325
04 A or B	7.5	2.133	325
05 A or B	8	2.000	325
06 A or B	9	1.778	325
07 A or B	10	1.600	325
08 A or B	11	1.455	325
09 A or B	12	1.333	325
10 A or B	12.6	1.270	325
11 A or B	13	1.231	325
12 A or B	15	1.067	325
13 A or B	16	1.000	325
14 A or B	18	0.889	325
15 A or B	20	0.800	325
16 A or B	22	0.727	325
17 A or B	24	0.667	325
18 A or B	26	0.615	325
19 A or B	28	0.571	325
20 A or B	30	0.533	325
21 A or B	33	0.485	325
22 A or B	36	0.444	325
23 A or B	39	0.410	325
24 A or B	43	0.372	325
25 A or B	47	0.340	325
26 A or B	51	0.314	325
27 A or B	56	0.286	325
28 A or B	62	0.258	325
29 A or B	68	0.236	325
30 A or B	75	0.213	325
31 A or B	82	0.195	325
32 A or B	91	0.176	325
33 A or B	100	0.160	325
34 A or B	110	0.145	325
35 A or B	115	0.139	325
36 A or B	120	0.133	325
37 A or B	130	0.123	325
38 A or B	150	0.106	325
39 A or B	180	0.089	325
40 A or B	200	0.080	440
41 A or B	220	0.073	440
42 A or B	240	0.067	440
43 A or B	270	0.059	580
44 A or B	300	0.053	580

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/37, MIL-PRF-27/38, MIL-PRF-27/39, MIL-PRF-27/40, MIL-PRF-27/41, MIL-PRF-27/42, MIL-PRF-27/43, MIL-PRF-27/45, MIL-PRF-27/46, MIL-PRF-27/47, MIL-STD-202-204, MIL-STD-202-211

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2020-016)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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